INCH-POUND

A-A-2937A
April 23, 2002
SUPERSEDING
A-A-2937
November 27, 1995

COMMERCIAL ITEM DESCRIPTION

REAMER, HAND, ADJUSTABLE

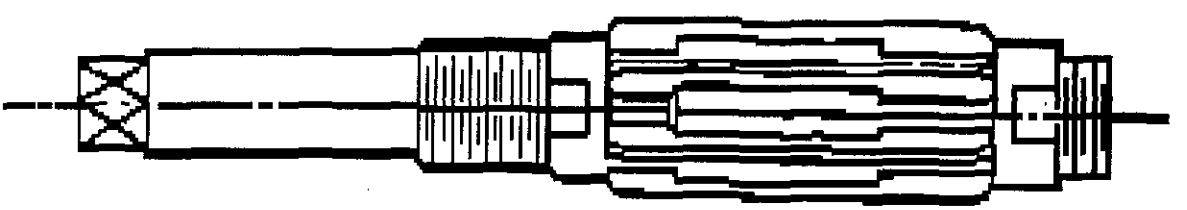
The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers adjustable hand reamers used in reaming hole diameters to specified tolerances.

2. CLASSIFICATION. *Not applicable*

3. SALIENT CHARACTERISTICS.

3.1 Design. The adjustable hand reamer shall be of the straight-fluted inserted blade type.



Adjustable, straight-fluted inserted blade reamer

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware SuperStore (6FEE), 1500 E. Bannister Rd Bldg. #6, Kansas City, Missouri 64131.

FSC 5110

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3.2 Dimensions and Sizes.

Nominal Size	Range of Expansion		Length of Blades *		Length Overall		Number of Blades
	Min. In.	Max In.	Min. In.	Max In.	Min In.	Max In.	Min.
8/A	¼	9/32	1-11/32	1-5/8	3-1/4	3-1/2	4
7/A	9/32	5/16	1-11/32	1/5/8	3-1/2	3-3/4	4
6/A	5/16	11/32	1-1/2	1-3/4	4	4-1/8	4
5/A	11/32	3/8	1-1/2	1-3/4	4-1/2	4-3/8	4
4/A	3/8	13/32	1-1/2	1-7/8	4-1/2	4-3/4	4
3/A	13/32	7/16	1-1/2	1-7/8	4-3/4	5	6
2/A	7/16	15/32	1-5/8	1-7/8	5	5-1/4	6
A	15/32	17/32	1-1/2	2	4	5-1/2	6
B	17/32	19/32	1-5/8	2-1/8	4-1/2	6	6
C	19/32	21/32	2	2-1/2	5	6-1/2	6
D	21/32	23/32	2	2-3/4	5-1/2	7	6
E	23/32	25/32	2-1/4	3	6	7-3/4	6
F	25/32	27/32	2-1/4	3-1/2	6-1/2	8-1/2	6
G	27/32	15/16	2-3/4	3-1/2	7	9	6
H	15/16	1-1/16	2-3/4	3-1/2	7-1/4	9-3/4	6
I	1-1/16	1-3/16	3	3-7/8	7-3/4	10-3/4	6
J	1-3/16	1-11/32	3-1/8	4	8	11-3/4	6
K**	1-11/32	1-17/32	3-3/8	4-1/2	9	12-3/4	6
	1-11/32	1-1/2	3-3/8	4-1/2	9	12-3/4	6
L**	1-17/32	1-25/32	3-1/2	4-1/2	13	14	6
	1-1/2	1-13-32	3-1/2	4-1/2	13	14	6
M**	1-25/32	2-3/32	4	5	15	16	6
	1-13/16	2-7/32	4	5	15	16	6
N**	2-3/32	2-15/32	4-13/16	5-3/4	17	18	6
	27/32	2-3/4	4-13/16	5-3/4	17	18	6
O**	2-15/32	2-31/32	5-3/16	6-1/2	19	20	6
	2-3/4	3-11/32	5-3/16	6-1/2	19	20	6

* = Length of blades to be measured between adjusting nuts with the reamer at maximum size
**= Either range is acceptable. However, when provided in sets, the ranges furnished shall be such that complete dimensional coverage is provided.

3.3 Materials. The body and adjusting nut part of the reamer shall be made of carbon steel and the cutting blades shall be made of high-speed steel.

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3.4 Construction. The reamer shall be fitted with removable cutting blades that shall be adjustable for reaming holes of any size within the range for which the reamer is designated. The reamer shall have straight, round shanks with square ends. The shanks shall be concentric with the cutting edges within 0.001" total indicator reading (tir) as measured between centers. The body portion of the reamer shall contain slots for holding the blades. The reamer shall be constructed to permit an adjustment of not less than 0.035" between the minimum and maximum size attainable with any one set of blades. The reamer assembly shall be capable of reaming round, straight and smooth holes free of chatter marks within any size range. The pilot bushing of the reamer shall have a surface roughness of not more that 25 micro-inches, arithmetical average (aa).

3.5 Blades. The blades shall be accurately machined and ground in sets to properly fit slots provided on the body of the reamer. Tolerances for the length of individual sets of blades shall not exceed plus or minus 0.0025". The width of the blades shall be uniform and interchangeable for the same size and make of reamer. Cutting surfaces of blades shall be concentric with the axis of the body in all positions throughout the range of adjustment. An adjusting nut, located at opposite ends of the blades, shall facilitate the adjustment of the blades. Blade ends shall be ground and rounded so that the blade ends do, not damage the adjusting nut, or washer when used.

3.6 Adjusting Nuts. Adjusting nuts shall have at least two flat sides, or other suitable means not requiring special tools or wrenches, for the expansion adjustment.

3.7 Hardness. The full length of the cutting edge of the reamer shall be hardened and tempered to a Rockwell "C" Scale value from 61-67 per ASTM E18.

3.8 Threads. The reamer shall have fine threads in accordance with FED-STD-H28.

3.9 Interchangeability. All part shall be designed and manufactured to definite standards, clearances and tolerances in order that any such parts of particular type or model may be replaced, interchanged and adjusted without modification of the replacement part or any other parts of the unit.

3.10 Identification Marking. The reamer shall be marked with the manufacture's name or identifying symbol, and the state or country of manufacture. All identification markings shall be engraved, etched, molded or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item. The reamer blades shall be marked in sets with the reamer's nominal size. In addition, the reamer shall be marked with the words "High Speed" or "High Speed Steel" or the corresponding abbreviation.

4. REGULATORY REQUIREMENTS.

4.1 Recovered Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal acquisition Regulation (FAR).

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch- pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawing, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for Inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

6. PACKAGING. Preservation, packaging and marking shall be as specified in the contract or order.

7. NOTES. This section contains information of a general or explanatory nature that is not mandatory.

7.1 Source of Documents.

American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959.

Federal Acquisition Regulation (FAR): Government Printing Office, Superintendent of Documents, Washington DC. 20401-9371.

General Services Administration, Specification Section (3FP-E), Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents.

a. Title, number, and date of Commercial Item Description.
b. Nominal size of reamer.
c. Packaging requirements.

7.3 National Stock Numbers (NSN's). A list of NSN's assigned that correspond to this CID. The list may not be indicative of all possible NSN's associated with the CID.

National Stock Number	Nominal Size	Range of Expansion		National Stock Number	Nominal Size	Range of Expansion	
		Min. In.	Max In.			Min. In.	Max In.
5110-00-529-2014	8/A	1/4	9/32	5110-00-289-1011	E	23/32	25/32
5110-00-529-2013	7/A	9/32	5/16	5110-00-260-1717	F	25/32	27/32
5110-00-541-4962	6/A	5/16	11/32	5110-00-260-1728	G	27/32	15/16
5110-00-260-1702	4/A	3/8	13/32	5110-00-260-1727	H	15/16	1-1/16
5110-00-541-4961	3/A	13/32	7/16	5110-00-232-6343	I	1-1/16	1-3/16
5110-00-720-4782	2/A	7/16	15/32	5110-00-260-1724	J	1-3/16	1-11/32
5110-00-186-6780	1A	7/16	1/2	5110-00-268-1235	K	1-11/32	1-17/32

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National Stock Number	Nominal Size	Range of Expansion		National Stock Number	Nominal Size	Range of Expansion	
		Min. In.	Max In.			Min. In.	Max In.
5110-00-260-1705	A	15/32	17/32	5110-00-232-6347	M	1-25/32	2-3/32
5110-00-260-1706	B	17/32	19/32	5110-00-720-4783	N	2-3/32	2-15/32
5110-00-260-1707	C	19/32	21/32	5110-00-720-4787	O	2-15/32	2-31/32
5110-00-260-1708	D	21/32	23/32	5110-00-915-0440		2-7/32	2-3/4
5110-00-596-1152	Kit containing sizes 4A-O						
5110-00-595-9283	Kit containing sizes A-H						
5110-00-947-3853	Kit containing sizes A-O						

7.4 Key words. Reamer, Hand, Adjustable

MILITARY INTERESTS:
NONE: DoD has no registered interest in revisions and amendments to this Commercial Item Description until further notice.

Preparing Activity:
GSA-FSS

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